SESCO CEMENT CORPORATION

Slag & GGBFS Comprehensive Market Dossier

Ground Granulated Blast Furnace Slag & Iron/Steel Slag
US Supply/Demand, SCM Markets & Trade Flow Analysis

CONFIDENTIAL — INTERNAL USE

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SLAG COMPREHENSIVE DOSSIER

Prepared: February 18, 2026 | **Version:** 1.0 | **Focus:** GGBFS & Iron/Steel Slag — US Supply/Demand, SCM Markets & Trade Flow Analysis

EXECUTIVE SUMMARY: Ground granulated blast furnace slag (GGBFS) is a supply-constrained, high-value supplementary cementitious material (SCM) facing a structural crisis. All available US GGBFS (~3–5 Mt/year) is consumed — the market is defined by supply limitations, not demand constraints. GGBFS represents less than 3% of total US slag tonnage but approximately 80% of its $600M value. The accelerating shift from blast furnaces to electric arc furnaces (EAFs now ~70% of US steelmaking) is progressively eliminating GGBFS feedstock, since EAF slag cannot substitute as an SCM. US import dependence is growing — Japan (52%), China (23%), and Brazil (11%) dominate. Meanwhile, carbon regulations (Buy Clean, LEED v5) are driving accelerating demand: GGBFS has 93–96% lower embodied CO2 than Portland cement (~35–67 vs. 844–967 kg CO2e/tonne). Supply is shrinking while demand is surging — this is the central tension in the slag market.

16 Mt
Total US Slag Sold (2024)
GGBFS <3% of tonnage

$600M
Total US Slag Value (2024)
GGBFS ~80% of value

3.7 Mt
Slag Cement Shipments (2024)
-16% from record 4.4 Mt (2023)

~96%
CO2 Reduction vs OPC
35 vs 967 kg CO2e/t

Executive Briefing — Strategic Implications for SESCO Cement

The GGBFS market presents a structural supply crisis that directly impacts SESCO's cementitious product portfolio and competitive positioning. This briefing highlights the five most critical takeaways:

- **Supply is shrinking permanently.** The US steel industry's shift from blast furnaces to electric arc furnaces (now ~70% of production) is eliminating GGBFS feedstock. EAF slag cannot substitute as an SCM. Only 4 integrated BF-BOF mills remain as primary domestic GGBFS sources, and further closures are planned.
- **Import dependence is rising.** US GGBFS imports reached ~2 Mt in 2024, with Japan (52%), China (23%), and Brazil (11%) as primary origins. Import prices are rising (Japan GGBFS at $120–160/Mt CIF), and global competition for limited BFS tonnage is intensifying.
- **Pricing power is extraordinary.** GGBFS prices have risen from $50–80/Mt (2018) to $100–140+/Mt (2024). As supply tightens, pricing is expected to continue rising. **SESCO opportunity:** GGBFS import and distribution represent a high-margin supply chain position.
- **Carbon regulations are accelerating demand.** GGBFS has 93–96% lower embodied CO2 than Portland cement. Buy Clean mandates, LEED v5 carbon limits, and state-level procurement policies are creating guaranteed demand growth even as supply contracts.
- **The SCM gap is widening.** Combined with fly ash decline (-64% since 2007 peak) and constrained GGBFS, the US faces a growing SCM deficit. Natural pozzolans and LC3 are emerging alternatives but cannot fully replace GGBFS performance in high-performance concrete. **SESCO opportunity:** Diversified SCM sourcing (slag + pozzolans) creates competitive advantage over single-source competitors.

1. TYPES OF SLAG

Type	Source	Key Properties	Primary Uses	SCM Value
GGBFS (Ground Granulated Blast Furnace Slag)	Blast furnace iron production; rapidly water-quenched, then ground	Latent hydraulic; >90% glass content; fineness 400–550 m²/kg; CaO 30–50%, SiO2 28–38%	Slag cement (Type IS); SCM in concrete (25–70% replacement); high-performance concrete	**HIGHEST — Premium SCM**
Air-Cooled BFS (ACBFS)	Blast furnace; slow air cooling	Hard, dense, angular aggregate; vesicular to glassy texture	Road base (42%), asphaltic concrete (17%), ready-mix agg (17%), fill (16%), RR ballast (7%)	LOW — Aggregate only
Steel Slag (BOF/EAF)	Basic oxygen furnace or electric arc furnace steelmaking	High free-CaO & iron oxide; volumetric instability (expansion); low hydraulicity	Road base (43%), fill (23%), RR ballast (11%), asphalt agg (10%), ice control	**NONE — Cannot substitute for GGBFS**
Copper Slag	Copper smelting; 1 ton per 3 tons Cu extracted	Fe2O3/SiO2 rich; limited pozzolanic character when vitrified	Abrasive blasting, aggregate, fill	MINIMAL
Phosphorus Slag	Phosphorus production	Lime-rich; can be granulated for cement use	Slag cement (limited regions)	MODERATE (niche)
Ferronickel Slag	Ferronickel production	Potential pozzolanic when vitrified; differs from BFS chemically	Alkali-activated cements (research stage), aggregate	EMERGING

CRITICAL DISTINCTION: Only GGBFS has significant cementitious (SCM) value. Steel slag from EAFs — which accounts for ~70% of US steelmaking — CANNOT substitute for GGBFS. Only ~1% of steel slag is used as a cement admixture vs. ~94% of GGBFS. This makes the blast-furnace-to-EAF transition the existential supply risk for the GGBFS market.

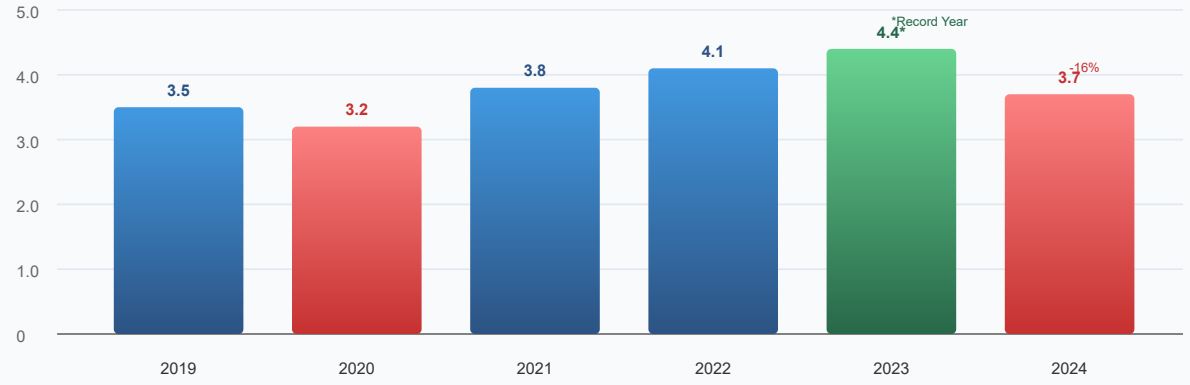
2. US PRODUCTION & DOMESTIC SUPPLY

2.1 Total US Slag Sales

Year	Slag Sold (Mt)	Value ($M)	BFS Share of Tonnage	GGBFS Share of Value
2021	17	~460	~50%	~85%
2022	15	~795	~52%	~88%
2023	16	~900	54%	~90%
2024 (est)	16	~600	N/A	~80%

2.2 Slag Cement Shipments (SCA Data)

US Slag Cement Shipments (Million Short Tons) — 2019 to 2024



Source: Slag Cement Association / Skyway Cement. Includes both domestic GGBFS production and imported slag ground domestically. 2024 decline driven by cement market softening and blast furnace idlings.

2.3 US Blast Furnace Infrastructure (2024–2025)

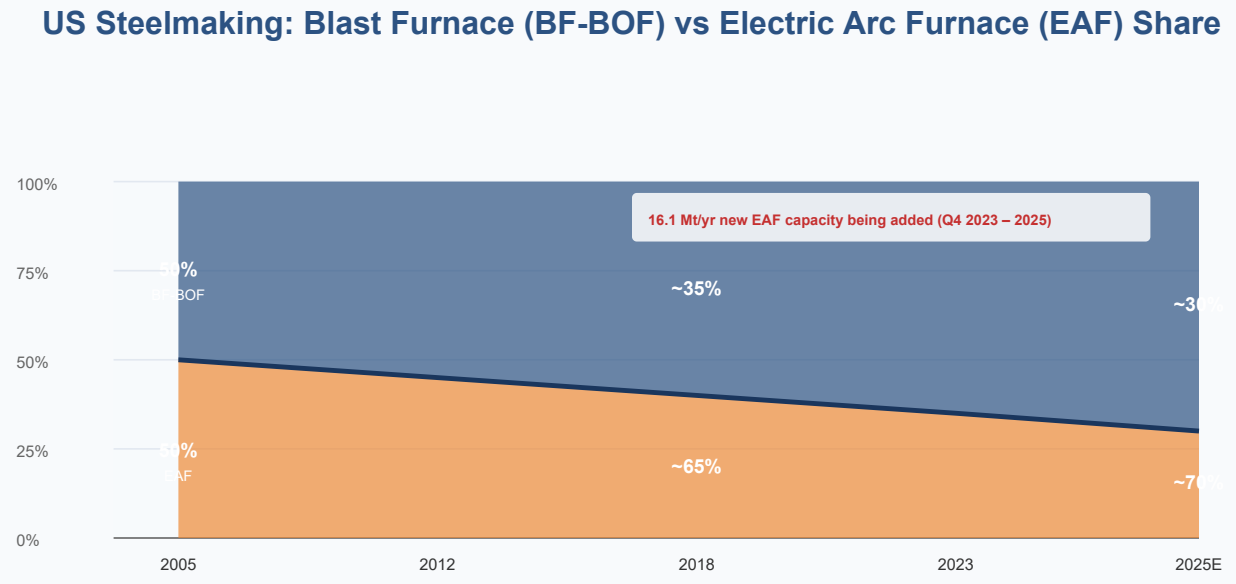
Company	Facility	Location	Blast Furnaces	Status	Granulation
Cleveland-Cliffs	Burns Harbor	Burns Harbor, IN	2 BFs (~5 Mt/yr)	Operating	New granulator Q4 2023; 2nd under construction
	Cleveland Works	Cleveland, OH	2 BFs (C-6 restarted mid-2025)	Operating	Existing
	Indiana Harbor	East Chicago, IN	3 BFs (1 operating)	Partial	Permits obtained for granulators
	Middletown Works	Middletown, OH	1 BF	Planned DRI conversion	$500M DOE funding for H2-DRI
	Dearborn Works	Dearborn, MI	1 BF	Idled (Mar 2025)	N/A
U.S. Steel / Nippon	Gary Works	Gary, IN	4 BFs (BF#14 reline 2025-26)	Operating	$3.1B investment through 2030
	Mon Valley (Edgar Thomson)	Braddock, PA	2 BFs	Operating	Existing

DOMESTIC GRANULATION CAPACITY: Only 4 active granulation operations as of 2024 (up from 2 in 2023). A new granulator started Q4 2023, with a second at the same mill expected fully operational in 2025. Permits obtained for additional granulators at another facility. Despite overall BF decline, investment in granulation is increasing because GGBFS is <3% of slag tonnage but ~80% of value.

2.4 Slag Processing Industry

- **25 companies** servicing active iron/steel facilities or reprocessing old slag piles
- **~120 processing plants** in **33 states**
- Key processors: Edw. C. Levy Co. (largest — 10+ Mt/yr), Charah Solutions, SATHI USA

3. THE EAF TRANSITION — EXISTENTIAL SUPPLY RISK



Sources: CATF, Argus Media, Fastmarkets. EAF share has grown from ~50% (2005) to ~70% (2025). EAF steelmaking produces NO granulated blast furnace slag suitable for use as an SCM. Only ~1% of steel/EAF slag is used as a cement admixture.

STRUCTURAL SUPPLY CRISIS: The US is adding 16.1 Mt/yr of new EAF steelmaking capacity (Q4 2023 through 2025) while blast furnaces are being idled and converted to DRI. Every BF closure permanently eliminates GGBFS feedstock. Cleveland-Cliffs' Middletown Works BF is planned for hydrogen-DRI conversion ($500M DOE funding). Dearborn Works BF was idled March 2025. The only bright spot: Nippon Steel committed $3.1B to reline 4 blast furnaces at Gary Works through 2030 — but this preserves existing capacity, it does not grow it.

3.1 Supply Impact Summary

Supply Threats

- **EAF expansion:** 16.1 Mt/yr new EAF capacity being added
- **BF idlings:** Dearborn idled (Mar 2025); Indiana Harbor partially idled
- **DRI conversion:** Middletown Works BF → H2-DRI ($500M DOE)
- **Auto demand weak:** Key BF-steel end market softening

Supply Stabilizers

- **Nippon Steel:** $3.1B Gary Works reline through 2030
- **Granulation expanding:** 2 → 4+ operations (2023-2025)
- **Burns Harbor:** 2nd granulator under construction
- **Indiana Harbor:** Permits for new granulators obtained

- **Only 7 integrated mills remain** in entire US
- **Imports growing:** Japan, China, Brazil filling gap

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4. GLOBAL PRODUCTION & MARKET SIZE

4.1 World Slag Production (2024 USGS)

Slag Type	Global Production (Mt)	Basis
Iron slag (blast furnace)	330–390	25–30% of crude pig iron production
Steel slag (BOF/EAF)	190–290	10–15% of raw steel production
Combined	**520–680**	**—**

4.2 Top Producing Countries

Country	Steel Production (Mt, 2023)	GGBFS Role	Key Detail
China	~1,020	~62% of global slag consumption (declining to ~55%)	Exports ~10 Mt GBFS annually
India	143	~16 Mt GGBFS consumption by 2025	Exports ~2.4 Mt slag
Japan	~87	~22 Mt GGBFS production; 15 Mt domestic use	Major exporter to US (52% share)
South Korea	~67	Slag = 35% of cement substitution	POSCO — global top-5 steelmaker
EU-27	~130	>90% utilization rate	Pioneer in ferrous slag recycling
US	~80	~3–5 Mt GGBFS consumed (all supply absorbed)	Only 7 integrated mills; growing importer

4.3 Global Market Size

Market Segment	2024 Value	Forecast	CAGR
Global GGBFS Market	$22–29 billion	$33–50B by 2031–33	4.4–6.3%
Global Slag Cement Market	$8.2–18.1 billion	$40.9B by 2034	8.6%
Global SCM Market	$91.1 billion	$119.9B by 2033	3.1%
US Slag Sales (USGS)	$600 million	—	—

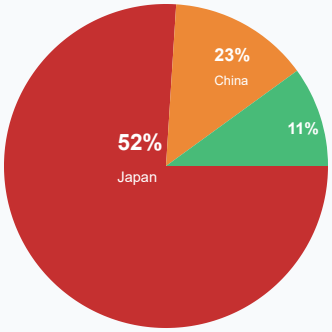
5. IMPORTS, EXPORTS & TRADE FLOWS

5.1 US Import Position — Growing Net Importer

Year	Import Volume (Mt)	Import Value ($M)	Notes
2018	2.6 (peak)	—	Record import volume
2019	2.08 (1.99 Mt GBFS/GGBFS)	—	—
2022	~2.0	$148	—
2023	~2.0+	$226	+53% value increase YoY

5.2 Import Source Countries (2020–2023)

US Granulated Slag Imports by Source Country



- Japan (52%) — Nippon Steel, other mills
- China (23%) — Baowu, others (tariff risk)
- Brazil (11%) — Declining share
- Mexico (5%) — Cross-border
- Other (9%) — Canada, S. Korea, EU

TREND:
Japan share grew from 40% → 52% (2019–2023)

Source: USGS MCS 2024-2025. HTS 2618.00.0000 (Granulated slag from iron/steel manufacture). Japan is the dominant and growing supplier.

5.3 Key Import Terminals

Terminal	Location	Operator	Details
Texas Lehigh Slag Facility	Houston Ship Channel, TX	Heidelberg Materials	New ~500K t/yr facility (completed Sep 2024)
Greens Port Industrial Terminal	Houston Ship Channel, TX	Watco / Charah Solutions	Slag grinding using patented tech ("MultiCem")
Titan America Terminal	Norfolk/Chesapeake, VA	Titan America	$37M upgrade with 70,000-ton dome storage
Cape Canaveral Terminal	Cape Canaveral, FL	Heidelberg Materials	Slag cement import/distribution
Various Gulf Coast	TX, LA, AL	Multiple	Major receiving region for imported GBFS

HTS CODE: 2618.00.0000 — Granulated slag (slag sand) from the manufacture of iron or steel. Also relevant: **2619** — Slag, dross (non-granulated) from iron/steel manufacture. Cement producers in Florida, California, Texas, Georgia, and the Carolinas are far from steel mills and rely heavily on imported GBFS.

6. PRICING

6.1 GGBFS Price Trend

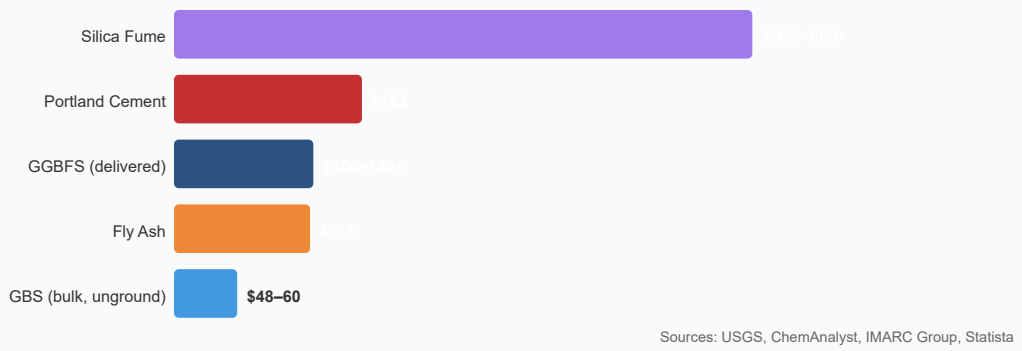
$100–140+
GGBFS Delivered/Ground (US)
Premium SCM product

$48–60
GBS Bulk/Unground (US)
Wholesale/import price

$163
Portland Cement (US, 2024)
Reference benchmark

$118
Fly Ash (US, 2025)
Competing SCM

6.2 Price Comparison — SCMs vs Portland Cement



6.3 Slag Value Paradox

Slag Product	% of US Slag Tonnage	% of US Slag Value	Price Range
GGBFS (ground)	<3%	~80%	$100–140+/ton
Air-Cooled BFS (aggregate)	~25%	~8%	$8–15/ton
Steel Slag (aggregate/fill)	~45%	~10%	$0–12/ton
Other/processing losses	~27%	~2%	Near-zero to negative

VALUE CONCENTRATION: GGBFS is one of the most value-concentrated industrial by-products in existence. Less than 3% of slag tonnage generates approximately 80% of the industry's total value. This extreme value differential is driving investment in granulation infrastructure even as the number of blast furnaces declines.

7. SCM MARKET CONTEXT

7.1 US SCM Supply Landscape

SCM Type	US Consumption (approx.)	Market Share	Supply Trend	Typical Cement Replacement
Fly Ash	~15–35 Mt	35–42%	Declining (coal plant closures)	15–35%
GGBFS	~3–5 Mt	~28%	Declining (BF closures/EAF shift)	25–50% (up to 70%)
Natural Pozzolans	Growing	~17%	Growing (volcanic, pumice)	15–25%
Silica Fume	<1 Mt	~12%*	Stable (niche)	5–10%
Calcined Clay / LC3	Minimal	Negligible	Emerging — US adoption very early	Up to 40% (LC3)

*Combined silica fume + metakaolin share

DUAL SCM SUPPLY CRISIS: Both of the US's primary SCMs — fly ash AND GGBFS — face structural supply declines driven by unrelated but concurrent industrial transitions. Fly ash supply is declining because coal-fired power plants are closing (-48% coal capacity since 2011). GGBFS supply is declining because blast furnaces are being replaced by EAFs (~70% of US steelmaking). There is no near-term large-scale domestic alternative. LC3 (limestone calcined clay cement) is positioned as the long-term replacement but US adoption is still minimal.

7.2 GGBFS Performance in Concrete

Advantages

- 25–50% cement replacement (up to 70% for durability, 95% in special applications)
- Superior long-term strength development
- Reduced permeability and porosity
- Greater resistance to chlorides, sulfates, and ASR
- Lower heat of hydration (ideal for mass concrete)
- Lighter color (aesthetic advantage)
- 93–96% lower embodied CO2 than OPC
- LEED v5 and Buy Clean compliance

Limitations

- Slower early strength gain (7-day)
- Requires activator (Portland cement) for hydration
- Supply-constrained — not always available
- Sensitive to temperature (cold weather curing challenges)
- Quality varies by source & granulation conditions
- Higher replacement rates require careful mix design
- Limited domestic supply (~3–5 Mt vs ~35 Mt fly ash)

8. MAJOR PRODUCERS & SUPPLIERS

8.1 Global GGBFS Market Leaders

Company	Country	GGBFS Market Share	Crude Steel (2024, Mt)	Role
China Baowu Steel	China	~6%	120+	Largest steelmaker globally
POSCO Holdings	South Korea	~5%	37.8	35% slag in Korean cement
Nippon Steel	Japan	~4%	43.6	Major US import supplier; Gary Works investor
ArcelorMittal	Luxembourg	~4%	65.0	Global integrated steel & slag
Tata Steel	India	~3%	30+	Growing Indian market

8.2 US Slag Cement Producers & Suppliers

Company	Role	Key Details
Skyway Cement (Eagle Materials)	Manufacturer	Chicago, IL. "100% US-sourced slag cement." Distributes via Eagle subsidiaries (Illinois Cement, Fairborn, Central Plains, Mountain, Texas Lehigh)
Heidelberg Materials / Lehigh Hanson	Producer & Importer	19 US cement plants. New Houston slag facility (~500K t/yr, Sep 2024). Cape Canaveral terminal.
Holcim / LafargeHolcim	Producer & Importer	13 US plants, 22 Mt/yr capacity. Slag-grinding in IL (~600K t/yr) and Midfield, AL (450K t/yr)
Edw. C. Levy Co.	Processor	Detroit, MI. Largest US slag processor (10+ Mt/yr). "LevyLite" GGBFS product. Proposed Green Circle grinding plant (315K t/yr)
Charah Solutions	Processor/Grinder	Greens Port, Houston. Patented grinding tech. "MultiCem" product.
SATHI USA	Importer/Supplier	Houston-based slag cement supplier
Argos USA	Producer	SCA member. Significant in Southeast US
St Marys Cement (Votorantim)	Producer	2 plants, 1.9 Mt/yr capacity
CEMEX	Producer	12 US plants, 15.7 Mt/yr capacity
CalPortland	Producer	Recent new SCA member

8.3 International Entrants (Notable)

Company	Country	Capacity	US Strategy
Ecocem	Ireland	2.4 Mt/yr (Europe's largest independent GGBS producer)	Attempted US entry via Orcem subsidiary (Vallejo, CA — planning denied). "ACT" product: 60% embodied carbon reduction.

9. END USES BY APPLICATION

9.1 GGBFS End Uses

Application	Share	Details
Cementitious materials (SCM)	~94%	Slag cement (Type IS), SCM in ready-mix concrete, precast
Other industrial	~6%	Soil stabilization, waste immobilization, geopolymers

9.2 Air-Cooled BFS End Uses

Application	Share
Road base and surfaces	39–42%
Asphaltic concrete aggregate	13–17%
Ready-mixed concrete aggregate	16–17%
Fill material	9–16%
Railroad ballast	7%
Other	Balance

9.3 Steel Slag End Uses

Application	Share
Road base	43%
Fill	23%
Railroad ballast	11%
Asphaltic concrete	10%
Ice control and other	13%

9.4 Other Slag Uses

- **Mineral wool:** ~60% of "other" slag category
- **Roofing:** Built-up and shingles (~15% of "other")
- **Glass manufacture:** ~5% of "other"
- **Agriculture:** Soil conditioning, fertilizer (calcium silicate)
- **Water treatment:** Filter media

10. PHYSICAL PROPERTIES & SPECIFICATIONS

10.1 GGBFS Chemical Composition

Oxide	Range (%)	Typical (%)	Comparison: Portland Cement
CaO	30–50	32–39	60–67
SiO2	28–38	33–37	17–25
Al2O3	8–24	10–13	3–8
MgO	1–18	7–10	0.5–6
Fe2O3	0.1–1.5	~1.0	0.5–6
SO3	0.1–2.0	~0.8	1.5–4.5

10.2 ASTM C989/C989M Grades

Grade	7-Day SAI (Avg/Min)	28-Day SAI (Avg/Min)	Application
Grade 80	Not specified	75% / 70%	Mass concrete (low heat of hydration)
Grade 100	75% / 70%	95% / 90%	General use; most similar to OPC performance
Grade 120	95% / 90%	115% / 110%	High-performance concrete; exceeds OPC at 28 days

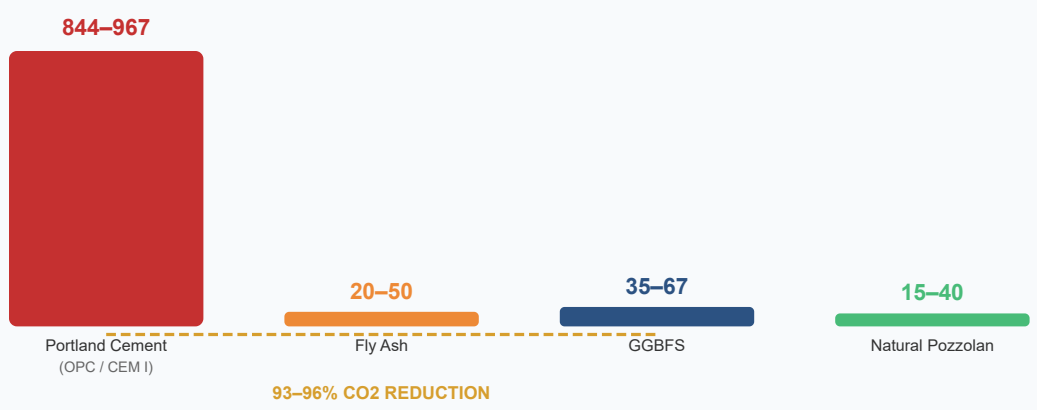
SAI = Slag Activity Index: compressive strength of slag mortar / reference cement mortar × 100

10.3 Physical Properties

Property	GGBFS	Portland Cement
Fineness (Blaine)	400–550 m²/kg	350–380 m²/kg
Specific Gravity	2.85–2.95	~3.15
Color	Off-white to beige	Gray
Glass Content	>90% (amorphous)	N/A (crystalline)
Particle Shape	Angular, irregular	Angular
Hydraulic Activity	Latent (requires activator)	Direct (self-hydrating)

11. CARBON & SUSTAINABILITY

11.1 Embodied Carbon Comparison



Sources: PCA, CSMA, AusLCI, various EPDs. GGBFS manufacture releases only ~35 kg CO2/tonne — energy used only for drying and grinding; melting energy allocated to iron production.

11.2 CO2 Reduction in Concrete

GGBFS Replacement Rate	Concrete CO2 Reduction	Strength Impact
25%	~25%	Comparable to OPC at 28 days
50%	~50%	Exceeds OPC at 56+ days
60%	~72% demonstrated	93.5 MPa achieved in studies
70%	~56–65%	Excellent durability applications
100% (alkali-activated)	~90%	71% of reference strength

CARBON VALUE PROPOSITION: Every ton of GGBS used instead of Portland cement saves approximately **650 kg CO2** in embodied emissions. Ecocem's "ACT" product has achieved ASTM-certified 60% embodied carbon reduction. With Buy Clean mandates and LEED v5 now requiring embodied carbon quantification, GGBFS is uniquely positioned as a premium low-carbon cementitious material — but supply cannot meet rising demand.

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12. REGULATORY & STANDARDS

12.1 Key Standards

Standard	Scope	Key Provisions
ASTM C989/C989M-22	Slag Cement for Concrete & Mortars	Defines Grades 80, 100, 120 by Slag Activity Index
ASTM C595/C595M-24	Blended Hydraulic Cements	Type IS (Portland-slag); allows up to 70% slag (newer: up to 95%)
ASTM C1157	Performance Hydraulic Cement	Any SCM proportion if performance criteria met
EN 15167-1	GGBS for Concrete (European)	European equivalent of C989

12.2 EPA & Beneficial Use

- EPA Comprehensive Procurement Guidelines (CPG) designate slag cement as a recommended product
- Slag classified as **non-hazardous industrial by-product**, not waste
- States with GGBFS specifications: AL, CT, FL, GA, IL, IN, MD, MI, NC, ND, OH, PA, SC, VA, WV, DC

12.3 Buy Clean & LEED v5

Regulation	Requirement	Impact on GGBFS
GSA Low-Embodied Carbon (LEC)	GWP limits for concrete; EPDs required for IRA-funded procurement	**Strong demand driver**
LEED v5	Mandatory embodied carbon prerequisite; EPDs required	**Strong demand driver**
State Buy Clean (CA, CO, NY, OR)	EPDs and/or embodied carbon limits for public works	**Growing demand driver**

REGULATORY TAILWIND: Since the GSA LEC program launch (2023), American manufacturers have published **14,000+ additional EPDs for concrete** (15% increase). LEED v5 requires products to outperform industry baseline by 20% or carry ISO 21930 "Optimized" tag. GGBFS at 93–96% lower embodied CO2 than OPC is the most impactful single ingredient for meeting these requirements.

13. LOGISTICS & HANDLING

13.1 Transportation Modes

Mode	Application	Details
Ocean Vessel	Import from Japan/China/Brazil	Handysize/Supramax (25K-60K DWT); loading ~8,000-10,000 t/day; freight ~$19-50/t (route/market dependent)
Rail	Inland distribution	Covered hopper cars from terminals & domestic mills; full-train preferred
Truck	Final delivery	Pneumatic tankers for ground product; dump trucks for aggregate slag; 100-200 mile radius
Barge	Waterway distribution	Mississippi system, Great Lakes, ICW; lowest cost/ton

13.2 Terminal & Storage Requirements

Requirement	Specification
Berth	Deep-water, capable of bulk vessel discharge
Unloading	Grab cranes or pneumatic; conveyor systems
GGBFS Storage	**Enclosed silos** (moisture-free); 70K-100K ton capacity typical
GBS Storage	Open storage possible with moisture management
Ground Product Handling	Pneumatic systems (250+ t/hr); enclosed to prevent moisture
Truck Loading	Weighbridge; pneumatic fill for tankers
Rail Access	Siding preferred for covered hopper loading
Quality Control	Segregation by type; chemistry documentation required for end-market compliance

HANDLING NOTE: Granulated slag is sandy and free-flowing when dry. Ground GGBFS is a fine powder requiring fully enclosed pneumatic handling. Moisture exposure degrades both products. Terminal infrastructure mirrors cement terminals — silos, pneumatic transfer, covered rail/truck loading — and can often share existing cement import infrastructure.

14. INDUSTRY TRENDS & OUTLOOK

US Slag/GGBFS Market — 5-Year Outlook (2025–2030)

Factor	Trend	Impact on Supply	Impact on Price
EAF Expansion	16.1 Mt/yr new EAF capacity; BF share declining to ~25%	**NEGATIVE**	**UP**
BF Closures/Idlings	Dearborn idled; Middletown → DRI; Indiana Harbor partial	**NEGATIVE**	**UP**
Gary Works Reline	Nippon Steel $3.1B through 2030; 4 BFs maintained	**STABILIZER**	NEUTRAL
Granulation Capacity	2 → 4+ operations; more permits obtained	**POSITIVE (modest)**	**DOWN (marginal)**
Import Growth	New terminals (Houston, Norfolk); Japan 52% share	**POSITIVE**	**DOWN (marginal)**
Buy Clean / LEED v5	Federal & state mandates; 14K+ new EPDs	N/A (demand side)	**UP**
Fly Ash Decline	Coal plants closing; fly ash supply -33% by 2030	**NEGATIVE (substitution pressure)**	**UP**
LC3 Alternative	Abundant raw materials; 30-40% CO2 reduction; early US stage	**LONG-TERM POSITIVE**	**DOWN (long-term)**
EAF Slag as SCM	Research ongoing; not commercially proven	UNCERTAIN	UNCERTAIN

NET ASSESSMENT: The US GGBFS market will remain severely supply-constrained through 2030. Domestic GGBFS feedstock will continue declining as blast furnaces close or convert to DRI/EAF. Imports (primarily from Japan) will partially fill the gap, supported by new terminal infrastructure in Houston, Norfolk, and Cape Canaveral. Carbon regulations (Buy Clean, LEED v5) are simultaneously accelerating demand for this 93-96% lower-carbon SCM. The result: **shrinking supply collides with surging demand** — supporting premium pricing and justifying investment in import infrastructure, granulation at remaining BFs, and alternative SCM development (LC3). GGBFS will remain one of the highest-value industrial by-products in the construction materials chain.

15. QUICK REFERENCE — KEY NUMBERS

Metric	Value	Trend
US Total Slag Sold (2024)	~16 Mt, $600M	Stable volume; value volatile
GGBFS Share of Tonnage	<3%	Tiny fraction
GGBFS Share of Value	~80% (~$480M)	Dominant value product
Slag Cement Shipments (2024)	3.7 Mt	-16% from 4.4 Mt record (2023)
US Blast Furnaces (operating)	~7 integrated mills, ~15 BFs	Declining (EAF shift)
EAF Share of US Steel	~70%	Rising (cannot produce GGBFS)
Domestic Granulation Operations	4 (up from 2 in 2023)	Expanding
US GGBFS Imports (2023)	~2 Mt, $226M	Value growing (+53% YoY)
Top Import Source	Japan (52%)	Share growing from 40%
GGBFS Price (delivered, ground)	$100–140+/ton	Premium SCM pricing
GBS Price (bulk, unground)	$48–60/ton	Wholesale/spot
Portland Cement Price	$163/ton	Reference benchmark
Fly Ash Price	$118/ton	Competing SCM; also supply-constrained
GGBFS Embodied CO2	35–67 kg CO2e/tonne	93–96% lower than OPC
Portland Cement Embodied CO2	844–967 kg CO2e/tonne	Industry baseline
CO2 Saved per Ton GGBS	~650 kg CO2	Key sustainability metric
Global GGBFS Market	$22–29 billion (2024)	$33–50B by 2031–33
Global Iron Slag Production	330–390 Mt	25–30% of pig iron output
Nippon Steel Gary Works Investment	$3.1 billion through 2030	4 BF relines; stabilizes domestic supply

SOURCES

Government: USGS Mineral Commodity Summaries 2024-2025 (Iron and Steel Slag); EPA Comprehensive Procurement Guidelines; FHWA Use of Air-Cooled Blast Furnace Slag; DOE Hydrogen Hub Funding
Standards: ASTM C989/C989M-22 (Slag Cement); ASTM C595/C595M-24 (Blended Cement); ASTM C1157 (Performance Cement); EN 15167-1 (GGBS, European)
Industry: Slag Cement Association (SCA); Skyway Cement Company; National Slag Association; Global Slag Conference 2024

Companies: Cleveland-Cliffs (operations, sustainability reports); U.S. Steel / Nippon Steel (Gary Works investment); Heidelberg Materials (Houston facility); Holcim (grinding stations); Edw. C. Levy Co.; Charah Solutions; Ecocem; SATHI USA
Market Research: Transparency Market Research (GGBFS Market); GM Insights (Slag Cement Market); CW Group (Global GGBFS Demand); ChemAnalyst (GGBFS Pricing); IMARC Group; Business Research Insights
Carbon & Sustainability: CSMA (Cementitious Slag Makers Association, UK); PCA; GSA Buy Clean / Low-Embodied Carbon Program; USGBC LEED v5; Climate Earth (EPDs); Ecocem ACT Certification
Steel Industry: CATF (US Steel Sector Decarbonization); Argus Media; Fastmarkets; World Steel Association
Trade Data: TrendEconomy (HTS 2618 trade flows); OEC (Granulated Slag trade); Datamyne
Academic: ScienceDirect (Blast Furnace Slag properties); Penn State (GGBFS technical guide); Purdue (EAF Slag reactivity research)

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